

EMERSON SALVADOR PLANCARTE CERECEDO

Backend Developer | .NET & Web Stacks

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PROFESSIONAL SUMMARY

Backend developer specializing in .NET (C#, ASP.NET) and web stacks (PHP, Node.js, Next.js/TypeScript), with SQL Server and PostgreSQL experience. Built and shipped data-integrated systems for a national electric utility, including a voltage-quality tool applied across 50+ localities. Additional experience in Android (Kotlin) and embedded systems (Raspberry Pi, Arduino, ESP32).

TECHNICAL SKILLS

Backend: C#, .NET Core/Framework, ASP.NET, PHP, Node.js, Laravel, Python, Java, REST APIs, LLM/RAG integration
Databases & Cloud: SQL Server, PostgreSQL (Supabase), MySQL, SQLite, Firebase, PI System (OSIsoft), Microsoft Azure, Vercel
Frontend: JavaScript, TypeScript, React, Next.js, jQuery, AJAX, Bootstrap, Tailwind CSS
Testing & DevOps: Automated Testing, CI/CD (GitHub Actions, automated Vercel deployments), Git
Mobile & Embedded: Kotlin, Android (Jetpack Compose, Room, Hilt), C/C++, Raspberry Pi, Arduino, ESP32

PROFESSIONAL EXPERIENCE

CFE – Guerrero-Morelos Transmission Zone (ZOTGM) Acapulco de Juárez, Guerrero, México
Contract Software Engineer (Ingeniero Contratista) Nov 2025 – Aug 2026

- Designed and delivered custom software for a 20+ person team, including a real-time grid load-capacity dashboard with PI System data integration (PHP/JavaScript), automated status alerts, and Excel/CSV export.
- Improved operational efficiency by 80% and reduced human error through intuitive interfaces.
- Delivered real-time visibility into voltage behavior via a local-first (on-premises) architecture, enabling immediate corrective action.

CFE – Guerrero-Morelos Transmission Zone (ZOTGM) Acapulco de Juárez, Guerrero, México
Software Development Intern (Residente) Apr 2024 – Apr 2025

- Designed and built a web application (.NET Framework 3.5, C#, PHP, JavaScript, jQuery, Bootstrap, AJAX) that integrates historical OSIsoft PI System data to automate Voltage Quality Index (ICV) evaluation under Mexico's Grid Code across 50+ localities, cutting analysis time by 80%+.
- Maintained 4 servers and 10+ workstations, resolved hardware/software failures, and ran updates and backups, sustaining 99%+ uptime; documented 100% of technical procedures.

PERSONAL PROJECTS

TecAssist.NET — RAG Technical Assistant (live) github.com/Emerson3101/TecAssistNet

- Rebuilt an award-winning university AI-assistant concept as a cloud-native RAG system: ASP.NET Core 10 Web API (Clean Architecture), EF Core, and Supabase (PostgreSQL + pgvector), with streaming, cited answers via NVIDIA NIM and a Next.js 16 chat UI.
- Secured the API with Supabase Auth (ES256 JWT/JWKS) and Postgres Row-Level Security; shipped 42 automated tests (xUnit, Testcontainers) via GitHub Actions CI/CD deploying to Azure App Service.

Wedding RSVP & Event Platform (Backend / Full-Stack) github.com/Emerson3101/wedproject

- Built and deployed a full-stack platform (Next.js 16, TypeScript, Supabase/PostgreSQL) for a real wedding with automated Vercel deployments: 20+ REST endpoints, cookie-based guest/admin auth behind an edge gate, and YouTube, Cloudinary, and Resend integrations.
- Designed a 7-table PostgreSQL schema (foreign keys, CHECK and partial unique constraints) with stored procedures for atomic RSVP upserts and idempotent voting; enforced seating capacity server-side (409 on overflow).

SharedCanvas — Real-Time Collaborative Drawing (Android) github.com/Emerson3101/SharedCanvas

- Built a two-user drawing app (Kotlin, Jetpack Compose, Clean Architecture, Hilt) with a dual-database sync engine: Firebase Realtime Database (~50–100ms live streaming) plus Cloud Firestore for durable storage.

MemeNote — Offline-First Productivity App (Android) github.com/Emerson3101/MemeNote

- Built an offline-first Android app (Kotlin, Jetpack Compose, Room, DataStore); cut the release APK from 26.8 MB to 6.5 MB with R8.

EDUCATION & AWARDS

Instituto Tecnológico Nacional de México, Campus Acapulco (TecNM Acapulco) Acapulco de Juárez, Guerrero, México
B.S., Computer Systems Engineering Jun 2020 – Dec 2025

- GPA: 92.33 / 100 | TOEFL English Certification | Winner, Local Technology Innovation Contest (Smart Cities), 2024